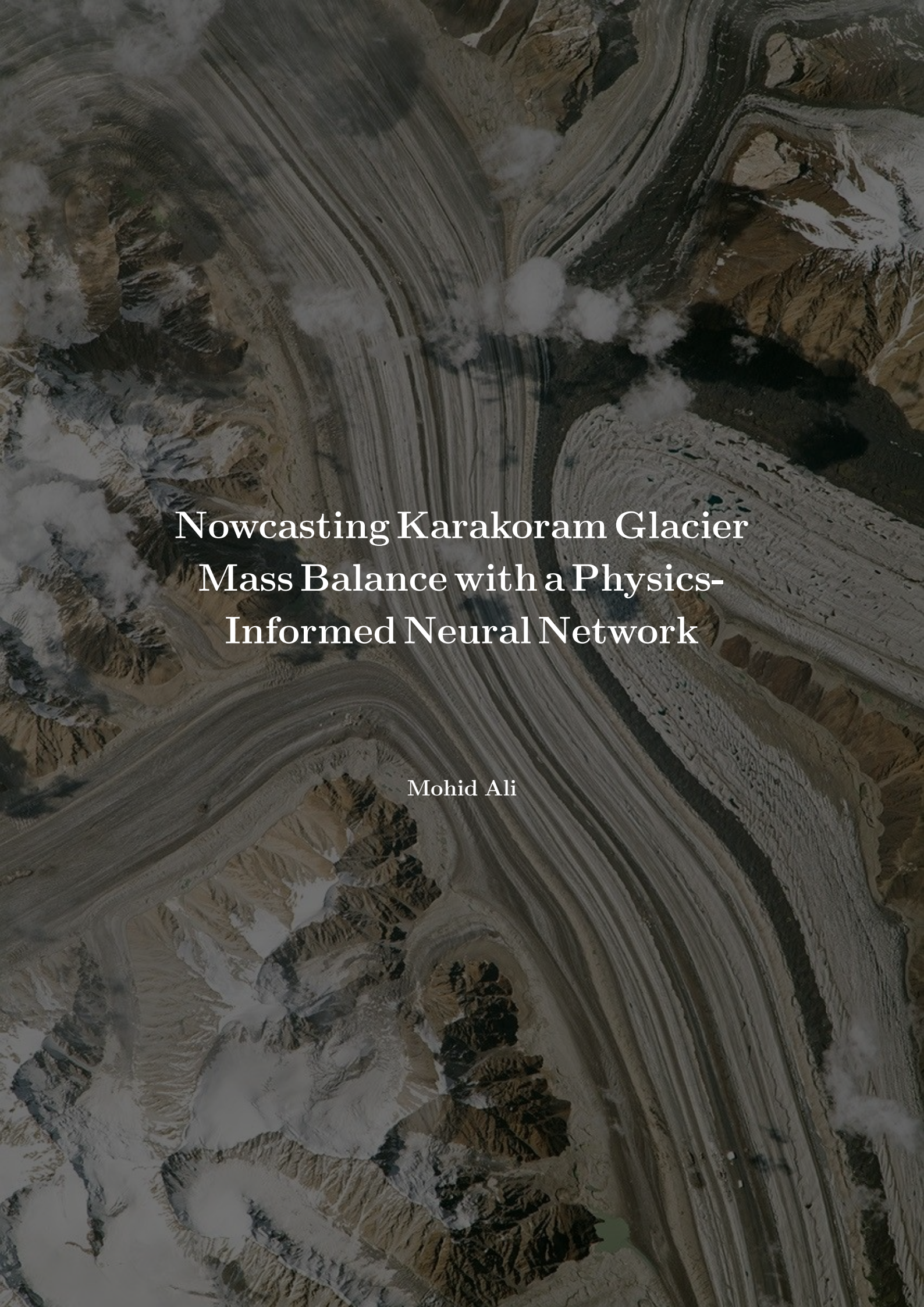


An aerial photograph of a vast glacier system, likely the Karakoram, showing intricate patterns of ice flow and sediment deposition. The image is overlaid with a dark, semi-transparent filter to enhance text readability.

Nowcasting Karakoram Glacier Mass Balance with a Physics- Informed Neural Network

Mohid Ali

Abstract

Most of the world’s glaciers are shrinking faster every decade, but the Karakoram range mostly is not; this decade-old puzzle, known as the Karakoram Anomaly, carries direct relevance for water security in the Indus basin. This project develops a physics-informed neural network (PINN) that nowcasts mass balance for five Karakoram and Western Himalaya glaciers – Baltoro, Biafo, Siachen, Pasu, and Batura – at 100 m elevation-band resolution, using satellite elevation change, climate reanalysis, terrain, and debris-cover data assembled through the Google Earth Engine platform. A preceding causal analysis, combining Granger causality, PC-algorithm causal discovery, and elevation-adjusted partial correlations, produces numeric outputs that feed directly into the PINN’s composite loss function to set its physics-term weights, so the link between the causal analysis and the physics loss is implemented in code instead of asserted in narrative. The model is validated under leave-one-glacier-out spatial holdout against a physical degree-day baseline and an XGBoost baseline, and an ablation study tests whether each physics term improves held-out accuracy. The physics-melt term is found not to clearly improve held-out accuracy, a result traced to the physical baseline it anchors to, while the conservation term against independent GRACE satellite gravimetry remains largely inert at its causally-derived weight. The PINN’s predicted profiles are nonetheless measurably smoother and more physically coherent than the alternatives, and the trained model is extended past the satellite record to produce an unvalidated nowcast of present-day mass balance, exposed through a live Streamlit dashboard.

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1 Introduction

Glacier mass balance, the net gain or loss of ice over a year, is the clearest signal of how a mountain range is responding to a warming climate. For the roughly 250 million people downstream of the Indus, it is a water security question as much as a scientific one. Global glaciers are losing mass at an accelerating rate, yet the glaciers of the Karakoram, straddling Pakistan’s northern border and including K2, have shown flat or even slightly positive mass balance for two decades. This Karakoram Anomaly has resisted a single clean explanation for over ten years, and little published work attempts to empirically ground a mass-balance model for this specific region, instead of applying textbook degree-day parameters calibrated for other glacierized regions unchanged.

This project addresses that gap with three questions: (1) what the anomaly actually looks like at fine, sub-glacier resolution across five representative glaciers, (2) what causally drives mass balance in this specific region, so the physics-informed loss function can be grounded in region-specific evidence, not textbook assumption, and (3) whether a model trained and validated only on the 2000–2020 window where satellite elevation-change data exists can be extended into an operational nowcast of the present. The result is a physics-informed neural network whose physics-term weights are mechanically derived from a preceding causal analysis, rigorously validated with an ablation study, and exposed through an interactive dashboard.

2 Study Area and Data

2.1 Study Area

Five glaciers were chosen to sit inside the anomaly itself, spanning the Karakoram and Western Himalaya and ranging from Pasu’s 62 km² to Siachen’s 1,088 km²: Baltoro and Biafo feed the Braldu and Shigar valleys near K2 and the Gasherbrum peaks, Siachen sits further east on the disputed India-Pakistan border, and Pasu and Batura lie in the Hunza valley to the northwest. The unit of analysis throughout the project is a 100 m elevation band per glacier, 230 bands in total after cleaning, since a single dh/dt number for an entire glacier hides almost everything interesting happening between a low, debris-covered tongue and a high accumulation zone.

Glacier	Latitude	Longitude	Area (km ²)
Baltoro	35.72	76.57	790.2
Biafo	35.94	75.75	532.0
Siachen	35.42	77.11	1088.0
Pasu	36.45	74.88	61.9
Batura	36.52	74.52	269.1

Table 1: Study glaciers, approximate coordinates, and RGI 7.0 area.

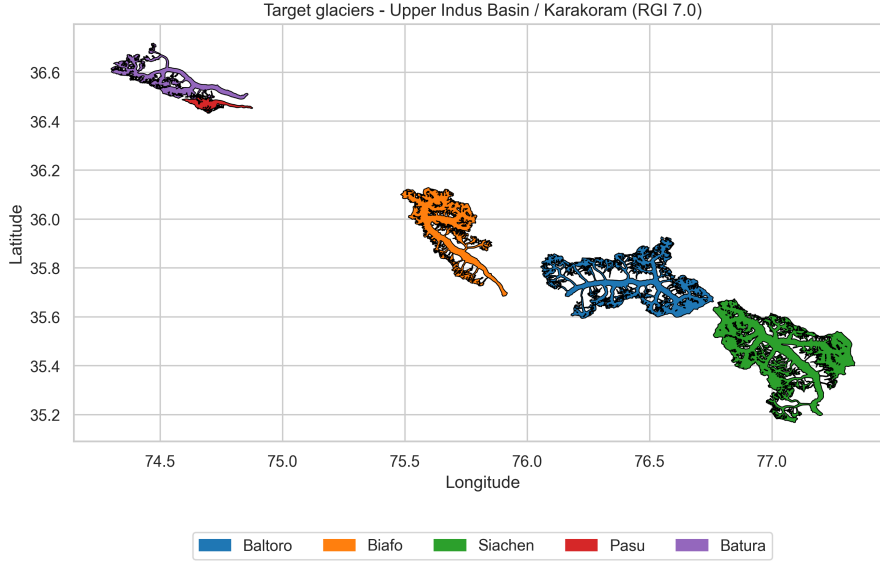


Figure 1: Outlines of the five study glaciers within the Karakoram and Western Himalaya (RGI 7.0).

2.2 Data Sources

All data is public and was assembled through the Google Earth Engine Python API directly to flat CSV, avoiding heavy local geospatial dependencies outside a single extraction notebook:

- **Randolph Glacier Inventory (RGI) 7.0:** glacier outlines, matched to each of the five target glaciers by `rgi_id` instead of name, since Passu is spelled “Pasu” in RGI and Siachen carries a local grid-reference code prefixed to its name.
- **Hugonnet et al. (2021) global glacier elevation-change dataset:** the primary dh/dt training target and its per-pixel uncertainty, filtered to the four independent, non-overlapping 5-year periods between 2000 and 2020.
- **GRACE / GRACE-FO satellite gravimetry (JPL mascons):** basin-scale total water storage trend for the Upper Indus, used as an independent cross-scale consistency constraint.
- **ERA5-Land reanalysis:** monthly and daily temperature, precipitation, snowfall, and snowmelt, downscaled per-pixel to true DEM elevation before computing degree days, since ERA5-Land’s own 11 km orography gives an effective lapse rate of only about 1 K/km if used raw.
- **NASADEM and HydroSHEDS:** terrain (elevation, slope, aspect) and the Upper Indus basin boundary used for the GRACE conservation constraint.
- **Scherler, Wulf & Gorelick (2018) global supraglacial debris-cover dataset:** published debris fractions, matched to the five glaciers by name against RGI 6.0 geometry and rasterized with Earth Engine.

2.3 Master Dataset Schema

All sources were merged into a single monthly panel at the (glacier, elevation band, month) level: 66,240 rows spanning 230 bands and 288 months (2000–2023), 22 columns. Static terrain and debris columns and the dh/dt target are broadcast across every month of a band; climate columns vary monthly. Key columns include:

- **dhdt_m_yr / dhdt_err_m_yr**: Hugonnet elevation-change rate and its per-pixel uncertainty (m/yr), populated only for the four independent 5-year periods
- **pdd_k_day**: positive degree-days, computed from the elevation-downscaled temperature
- **t2m_ds_c**: downscaled monthly mean temperature (°C)
- **precip_mm, snowfall_mm, snowmelt_mm, swe_mm**: monthly climate totals from ERA5-Land, with **swe_mm** masked above 3,900 m where it saturates at ERA5-Land’s 10 m water-equivalent ceiling
- **debris_frac**: fraction of a band’s area under supraglacial debris, from the Scherler dataset
- **z_mean, slope_deg, aspect_sin, aspect_cos**: terrain descriptors per band
- **area_km2**: band area, used for area-weighting every glacier-level statistic

2.4 Data Preprocessing

Six degenerate bands with area under 0.01 km² and zero clear pixels were dropped. The 0.01 km² threshold falls in a real gap between the smallest retained band (0.0101 km²) and the largest dropped one, so it reflects an actual break in the data instead of an arbitrary line. **ndsi_mean** was clipped to its physical [-1, 1] range in the four bands where Landsat surface reflectance goes slightly negative in deep shadow. Only the four independent, non-overlapping 5-year Hugonnet periods were merged into the master table; three longer aggregate windows built from the same underlying observations were deliberately left out to avoid leakage. No missing value was imputed: **swe_mm** is missing above 3,900 m by design, and **dhdt_m_yr** is missing for every month from 2020 onward, since Hugonnet’s satellite record stops there while the climate features keep extending, which is precisely the nowcast window this project is built to fill.

3 Methodology

3.1 Exploratory and Geospatial Analysis

Univariate and multivariate exploratory analysis established two conventions used throughout the rest of the project: every glacier-level statistic is area-weighted by **area_km2**, since a plain per-band mean over-weights small high-elevation slivers relative to the tongue and reads meaningfully more negative than the true signal, and dh/dt trends are always examined at elevation-band resolution, since a single glacier-wide number erases exactly the variation that matters. This phase also produced five concrete hypotheses handed to the

causal phase, the most important being that debris cover’s raw correlation with dh/dt is roughly half confounded by elevation, and that a basin-wide melt-energy lead on GRACE storage looked real in a simple lag correlation but needed a formal causal test.

3.2 Causal Inference

This is the project’s central methodological contribution: a preceding causal analysis phase feeds its literal numeric outputs directly into the neural network’s loss function, turning causal findings into loss weights. Granger causality tests (statsmodels) on the raw basin-climate and GRACE series found every climate variable Granger-causing GRACE *and* GRACE Granger-causing every climate variable, the classic signature of two series that simply share an annual cycle. After deseasonalizing (subtracting each calendar month’s own mean) and retesting, only precipitation retained a significant, if still bidirectional, relationship; positive degree-days, snowmelt, and temperature all lost significance entirely. A PC-algorithm causal discovery pass (`causal-learn`) independently confirmed a direct edge from debris cover into dh/dt , and elevation-adjusted partial correlations quantified the surviving, non-confounded relationships: debris cover and melt energy are the two variables with a genuine, non-elevation-confounded relationship to mass balance, while temperature’s own partial relationship flips sign once elevation is controlled for, revealing it to be largely an elevation proxy in this dataset. A counterfactual-style, elevation-matched comparison further found that debris cover’s effect on dh/dt reverses sign with elevation, insulating the ice at 3,000 m but coinciding with faster thinning at 3,500–4,000 m, a non-monotonic pattern that should not be averaged away as noise.

3.3 Baseline Predictive Modeling

Two baselines were built under leave-one-glacier-out spatial holdout: a plain temperature-index (degree-day) physical model with no fitting, using literature melt-factor constants for bare ice and debris-covered ice, and an XGBoost model trained on eleven climate, terrain, and debris features. Both were compared against a naive training-mean baseline to test whether either one reliably generalizes to a glacier it has never seen.

3.4 Physics-Informed Neural Network

The PINN is a compact feedforward network trained with a composite loss, $L = L_{data} + \lambda_1 L_{physics} + \lambda_2 L_{conservation}$. L_{data} is weighted by each measurement’s own Hugonnet uncertainty, since that uncertainty grows from a few meters per year near the tongue to over 15 m/yr at extreme elevations against a true signal well under 0.5 m/yr, and an unweighted loss would mostly fit noise across the least-observed majority of glacier area. $L_{physics}$ pulls predictions toward the degree-day baseline’s melt-driven shape, and $L_{conservation}$ pulls the model’s area-weighted aggregate prediction toward consistency with GRACE’s independently measured basin-wide trend, scaled by the fraction of the Upper Indus basin’s water-storage signal that the five study glaciers’ own historical area-weighted dh/dt can plausibly explain (about 6.2 percent). Critically, λ_1 and λ_2 are not hand-picked: $\lambda_1 = 0.166$ is the absolute value of positive degree-days’ elevation-adjusted partial correlation with dh/dt from the causal phase, and $\lambda_2 = 0.099$ is derived from precipitation’s deseasonalized Granger p-value against GRACE, kept deliberately modest because that relationship only barely survived deseasonalizing in the first place.

3.5 Training, Tuning and Validation

A full hyperparameter sweep across network depth, learning rate, and physics-weight scale was run under the same leave-one-glacier-out scheme, each of the twelve combinations scored by its mean RMSE across all five glaciers together, since a config that happened to fit one glacier well could just be measuring luck. An explicit ablation study then retrained the model with each physics term individually zeroed out, and a five-seed ensemble quantified run-to-run variance from initialization and batch order alone.

3.6 Nowcast Generation

Every validation number above only ever tests the model on 2000–2020, the window where Hugonnet satellite coverage exists. To actually nowcast, the tuned model was run on real ERA5-Land climate data covering the years past that satellite record, using each glacier’s own held-out standardization. One non-obvious correction was required: since the model was trained on positive degree-days summed over fixed five-year periods, degree-days over the shorter, still-open nowcast window were rescaled to the same five-year footing before inference, avoiding a spurious low-melt-energy artifact from the window’s shorter length.

4 Evaluation Metrics

All predictive models, physical, XGBoost, and PINN alike, were evaluated under leave-one-glacier-out spatial holdout: for each of the five glaciers in turn, a model is trained on the other four and tested on the one held out completely. A random split would let elevation bands from the same glacier leak between train and test, since neighboring bands on one glacier are highly spatially correlated, so this is the only valid way to test whether a model generalizes to a glacier it has never seen. RMSE and bias were computed per glacier on the four independent Hugonnet periods. Ablation was evaluated the same way, retraining each loss-term variant under the identical holdout scheme and comparing mean RMSE across all five glaciers.

Causal claims cannot be evaluated with a train/test split, since there is no ground-truth counterfactual to check against. Confidence instead rests on a refutation-style check built directly into the methodology: deseasonalizing before re-running the Granger tests functions as a refutation test, and it is exactly what exposed the melt-energy-leads-GRACE finding as a seasonal artifact once the seasonal cycle was subtracted out, so a result that looked clean under a naive test never made it into the final causal claims.

5 Results Summary

5.1 Predictive Model Performance

Even before any modeling, the raw area-weighted dh/dt signal shows the anomaly directly: close to flat for three of the five glaciers, with only Baltoro and Pasu unambiguously losing mass, against a global glacier average of roughly -0.5 to -1.0 m/yr.

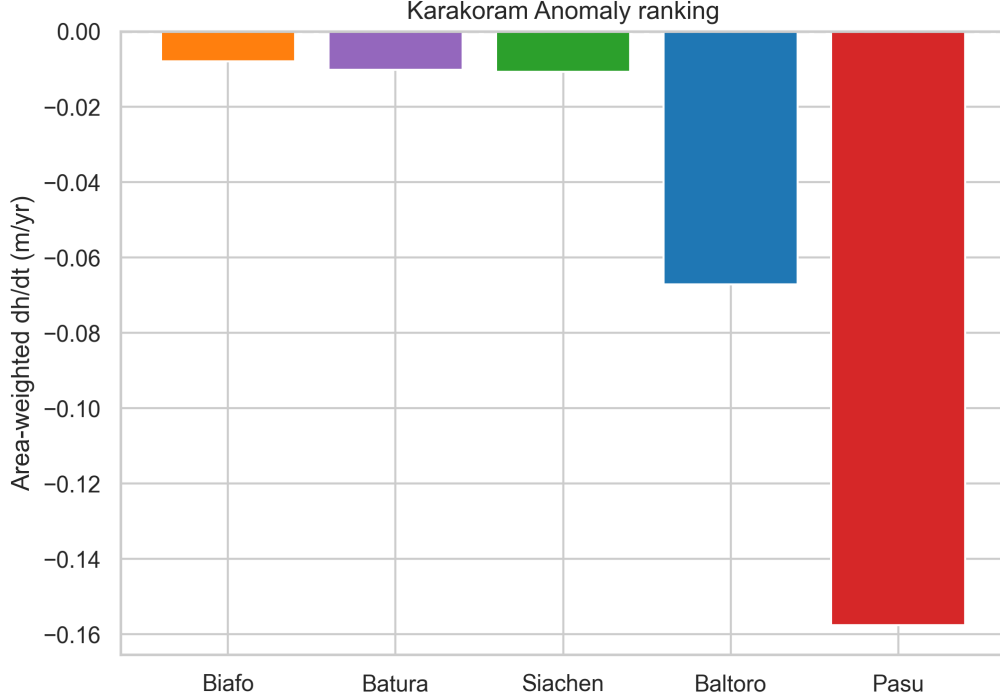


Figure 2: Area-weighted dh/dt ranking across the five glaciers, 2000–2020. Only Baltoro and Pasu show unambiguous mass loss.

Under leave-one-glacier-out holdout, the plain physical model is worse than simply guessing the training-set mean on every single glacier, and the tuned PINN clearly improves on the physical baseline everywhere, though it does not surpass XGBoost on raw RMSE on any glacier.

Glacier	Naive Mean	Physical (DDF)	XGBoost	PINN (tuned)
Baltoro	0.46	0.80	0.53	0.71
Biafo	0.57	1.10	0.86	0.91
Siachen	0.95	1.17	0.78	0.92
Pasu	1.04	1.93	0.96	1.13
Batura	0.58	1.57	0.53	0.87

Table 2: Leave-one-glacier-out RMSE (m/yr) for all four models. Neither baseline reliably generalizes on its own, providing empirical motivation for the physics-informed approach.

The physical model’s failure is concentrated at low elevation. At Pasu, for example, it predicts melt down to roughly -6 m/yr at the glacier tongue against an observed -1 m/yr, a systematic overshoot visible directly in the elevation profile below, while XGBoost tracks the observed profile far more closely across the full elevation range.

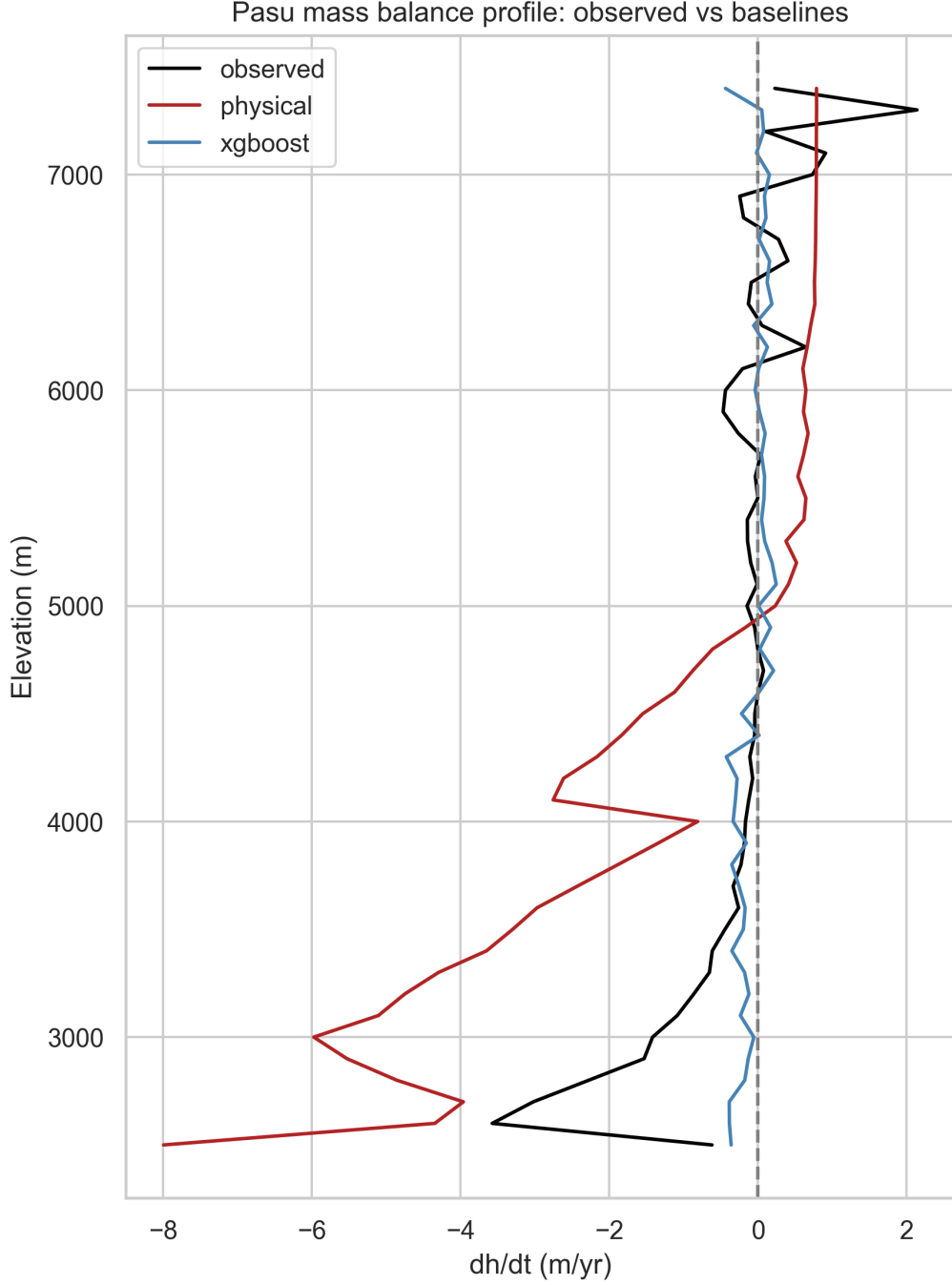


Figure 3: Observed versus predicted elevation profile for Pasu, all three baselines. The physical model overshoots melt sharply at low elevation.

5.2 Causal Findings

Elevation-adjusted partial correlations isolate debris cover and melt energy as the two variables with a genuine, non-confounded relationship to dh/dt (raw to partial: debris cover -0.46 to -0.24, positive degree-days -0.42 to -0.17), while temperature’s own partial relationship flips sign once elevation is controlled for. These partial correlations were computed using an NDSI-based debris-cover proxy and have not been recomputed with the Scherler debris-cover dataset described in Section 2.2; they should be read as indicative of the underlying relationship, not as exact under the debris measure used elsewhere in

this report.

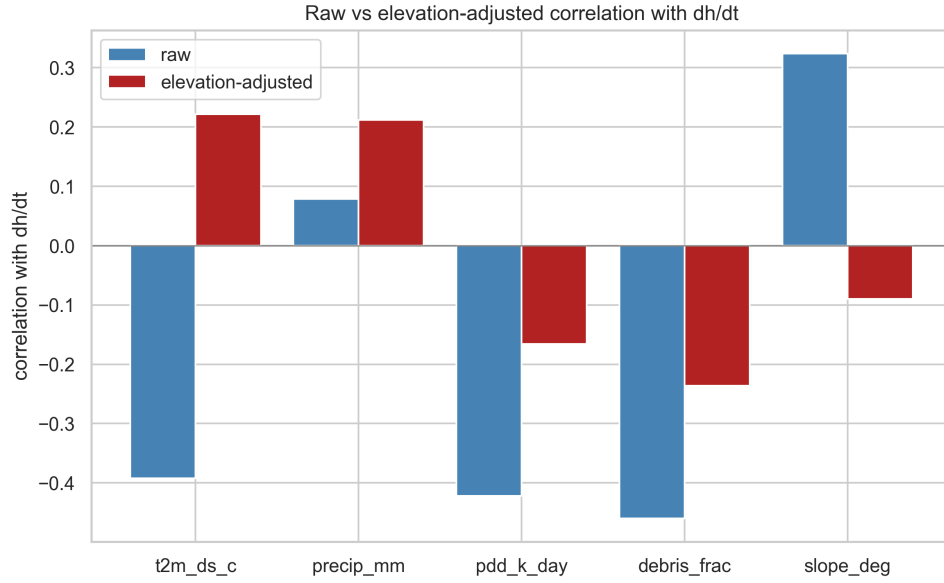


Figure 4: Elevation-adjusted partial correlations with dh/dt , raw versus partial, for each candidate driver.

A counterfactual-style, elevation-matched comparison shows that debris cover's effect on dh/dt is not constant with elevation: at 3,000 m, heavily debris-covered bands thin less than clean-ice bands at the same elevation (-0.72 vs -0.93 m/yr), consistent with debris insulating the ice, but at 3,500 m and 4,000 m the effect reverses, with debris-covered bands thinning more (-0.72 vs -0.27 m/yr and -0.49 vs -0.10 m/yr). A single, elevation-independent debris coefficient would misrepresent this relationship; the physics loss described in Section 3.4 does not include a debris-specific term, so this interaction is not directly encoded in the current model.

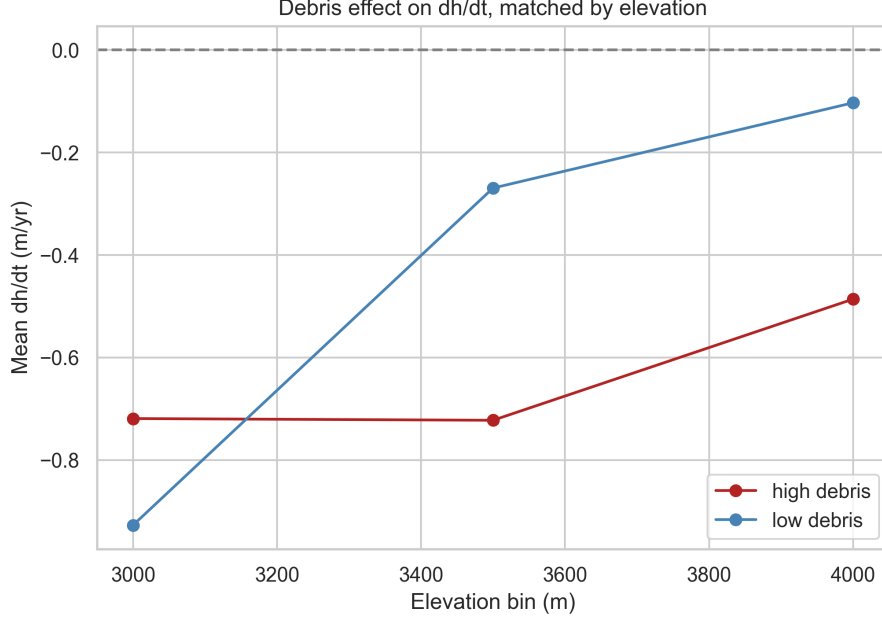


Figure 5: Mean dh/dt for high- versus low-debris bands, matched by elevation. Debris cover’s apparent effect reverses sign between 3,000 m and 3,500–4,000 m.

5.3 Physics-Informed Model and Ablation

The ablation result is the most consequential finding of this study: the best average held-out RMSE among the four loss-term variants tested comes from removing the physics-melt term entirely, while the conservation term produces almost no change in either direction.

Loss variant	Mean RMSE across 5 glaciers (m/yr)
Data loss only	0.858
No physics-melt term	0.875
No conservation term	1.016
Full composite loss	1.021

Table 3: Held-out RMSE for each ablated loss-term variant, per glacier. The physics-melt term does not clearly earn its place on RMSE; the conservation term is close to inert.

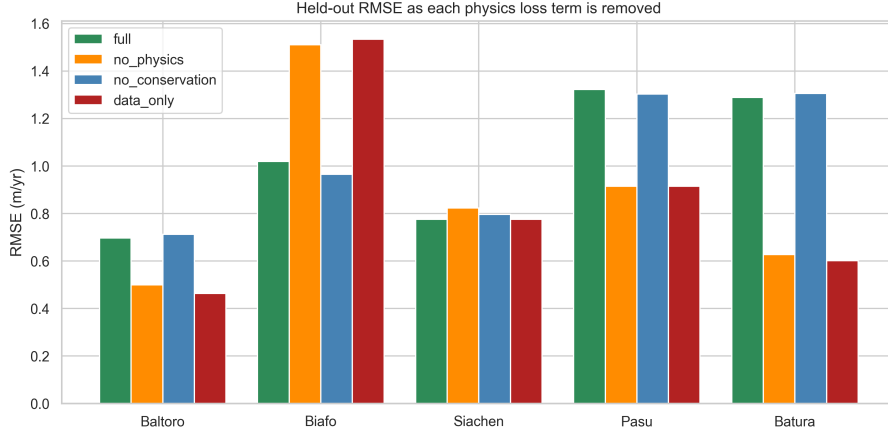


Figure 6: Held-out RMSE by ablated loss variant, per glacier.

This traces back directly to the baseline result above: the physics term anchors predictions to the same degree-day baseline already shown to be worse than a naive mean on every glacier, so anchoring to it is net detrimental more often than it is beneficial. This pattern holds under both the NDSI-based debris proxy and the Scherler debris-cover dataset used in the final pipeline, indicating the effect is not an artifact of the debris-cover measure. What the physics-informed formulation does provide, independent of this accuracy gap, is smoother, more physically coherent output: mean roughness between adjacent 100 m bands is 0.140 m/yr for the PINN against 0.153 for XGBoost and 0.196 for the physical baseline, with the PINN smoother than XGBoost on four of the five glaciers.

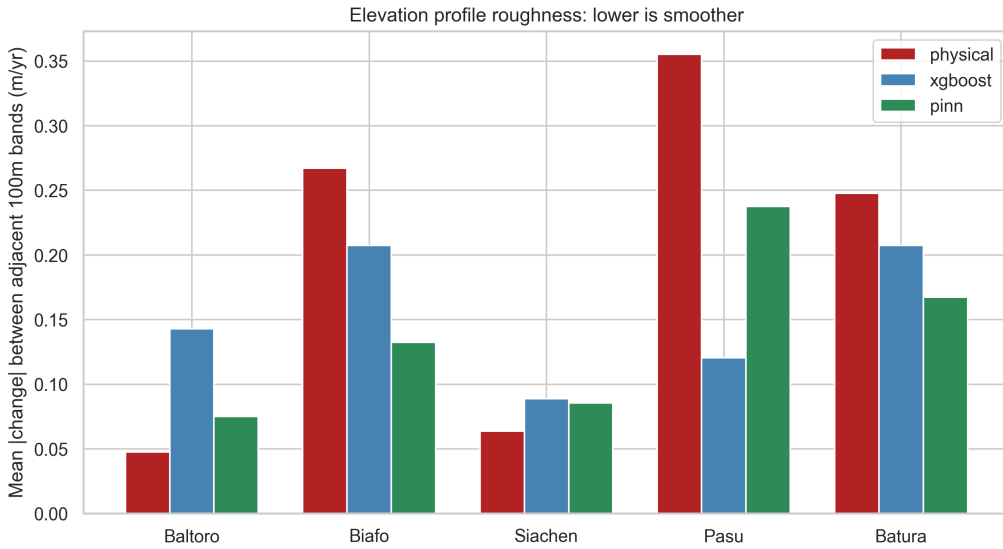


Figure 7: Mean roughness between adjacent 100 m elevation bands, by model and glacier. Lower is smoother.

An independent check against GRACE’s basin-scale trend confirms that λ_1 , not λ_2 , drives the model’s basin-scale behavior. The implied basin-wide water-storage trend is 0.268 cm/yr under the full composite loss and 0.291 cm/yr with the conservation term removed, both roughly an order of magnitude from the calibrated target of -0.029 cm/yr and barely distinguishable from each other. Removing the physics-melt term instead, with

the conservation term still active, brings the implied trend to 0.034 cm/yr, far closer to the target.

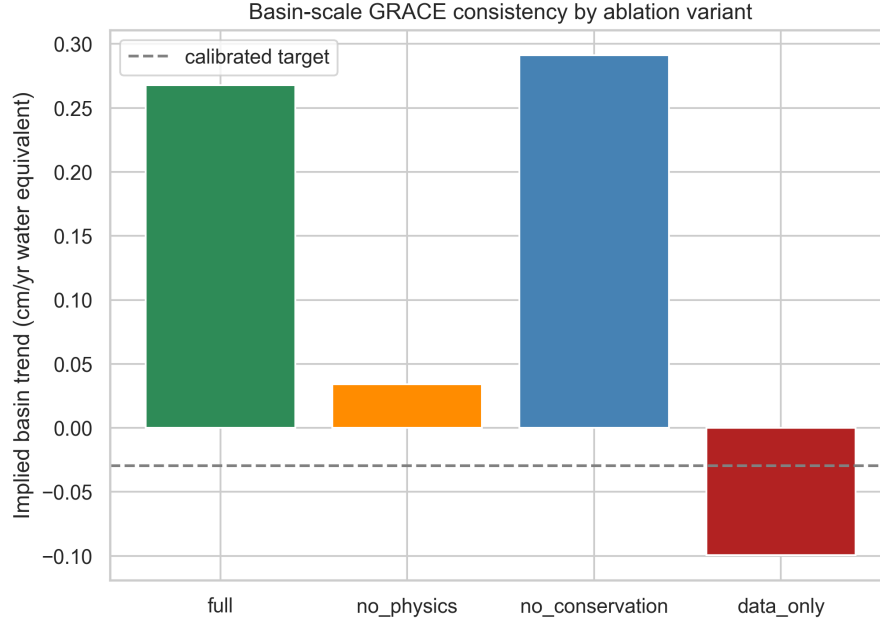


Figure 8: Implied basin-wide water-storage trend under each ablation variant, against the calibrated GRACE target.

5.4 Nowcast Results

Extending the tuned model past the satellite record using current ERA5-Land climate data produces an unvalidated nowcast of present-day mass balance.

Glacier	Nowcast, 2020-01 to 2026-07 (m/yr)	2000–2020 observed average (m/yr)
Baltoro	+0.34	-0.07
Biafo	-0.06	-0.002
Siachen	+0.24	+0.002
Pasu	-1.19	-0.15
Batura	-0.87	+0.002

Table 4: Area-weighted PINN nowcast against the 2000–2020 observed baseline, per glacier.

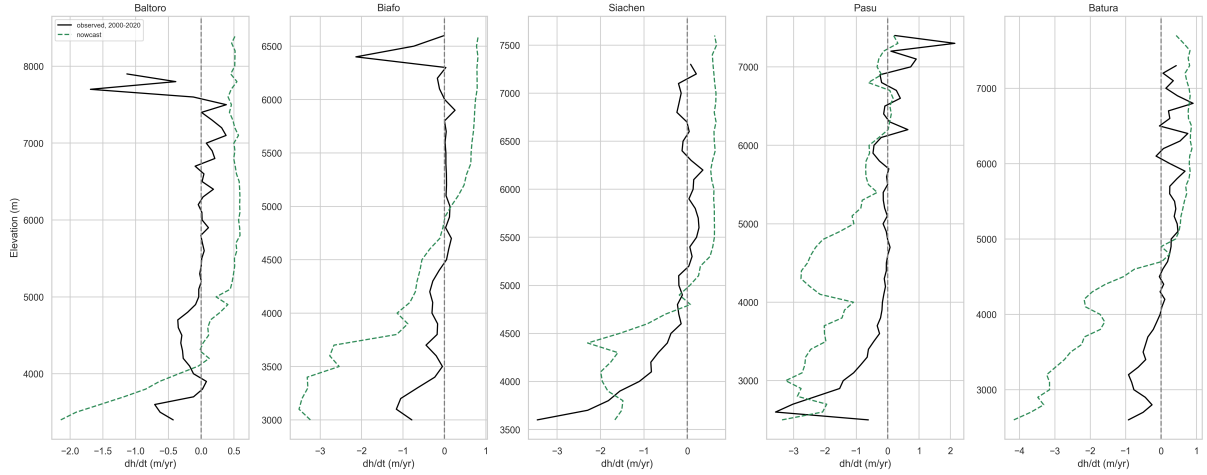


Figure 9: Elevation profile, 2000–2020 observed versus the 2020-01 to 2026-07 PINN nowcast, all five glaciers.

Baltoro and Siachen’s positive nowcasts likely reflect the same positive prediction bias observed in the historical holdout and should be interpreted with caution. Batura’s shift from near-flat over the historical record to strongly negative in the nowcast window is the one unexplained signal in this result; it remains unvalidated, since no per-glacier satellite coverage exists for this period.

6 Model Interpretability

Interpretability here works differently for the causal component than for the neural network. The PC-algorithm causal graph is an explicit, human-readable structural model, so every edge, and every physics-loss weight ultimately derived from it, can be traced back to a specific, inspectable statistical test. Several edges orient in physically counterintuitive directions, a known limitation of conditional-independence-based causal discovery under collinearity, so edges are read here as “these variables are directly related” instead of as proven causal arrows.

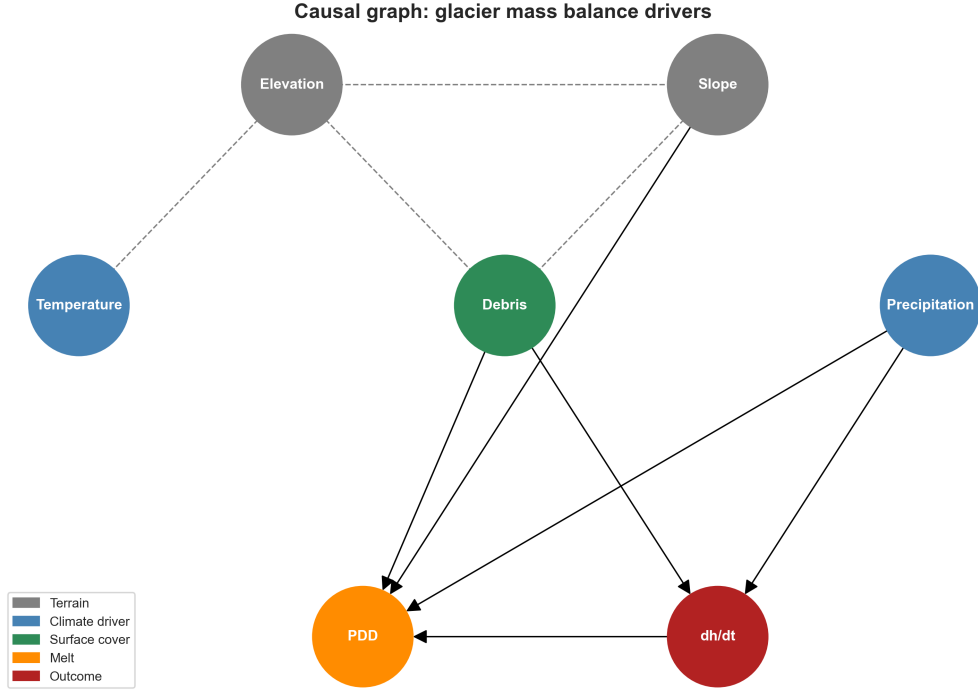


Figure 10: PC-algorithm causal graph over the period-level variable set, the structural basis for the physics-loss weighting.

The PINN itself skipped a post-hoc feature-attribution pass; its interpretability is built into its construction instead, since λ_1 and λ_2 are named, traceable numbers from the causal graph above. Its elevation profiles were also examined as an implicit physical-plausibility check: they are measurably smoother, band-to-band, than either baseline, indicating that the physics anchoring imposes structure on the output even where it does not improve RMSE.

7 Streamlit Dashboard

The full pipeline, elevation-band profiles for all five glaciers, per-band climate detail, and the nowcast itself, is exposed through a live, interactive dashboard built with Streamlit. The dashboard is deployed at <https://glacier-mass-balance-nowcaster.streamlit.app> and is publicly accessible. It is deliberately PINN-only, leaving out every baseline model discussed in this report, so the tool stays focused and easy to use. It is organized into three pages, described below.

7.1 Glacier Overview

The landing page reports a glacier’s key metrics, area-weighted dh/dt , debris cover, total area, and elevation-band count, alongside its elevation profile and a locator map, with a sidebar selector that drives every page in the app.

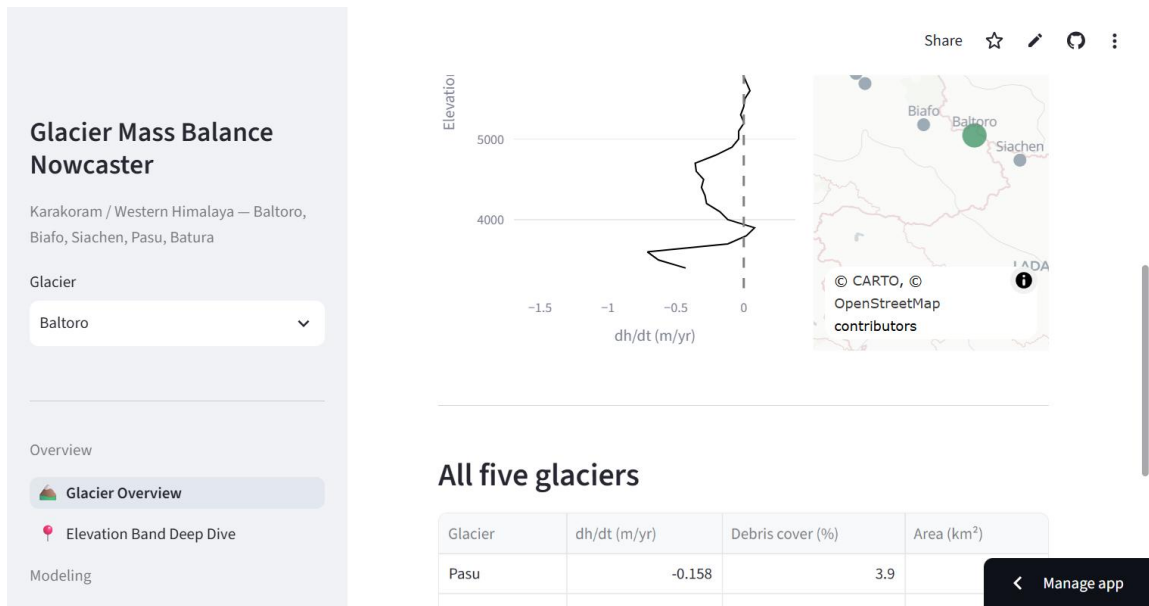


Figure 11: Glacier Overview page, showing Baltoro’s key metrics and elevation profile.

7.2 Elevation Band Deep Dive

Selecting an individual elevation band reveals its area, debris cover, slope, and mean dh/dt , alongside its monthly climate trend, temperature and positive degree-days plotted together over the full 2000-to-present record.

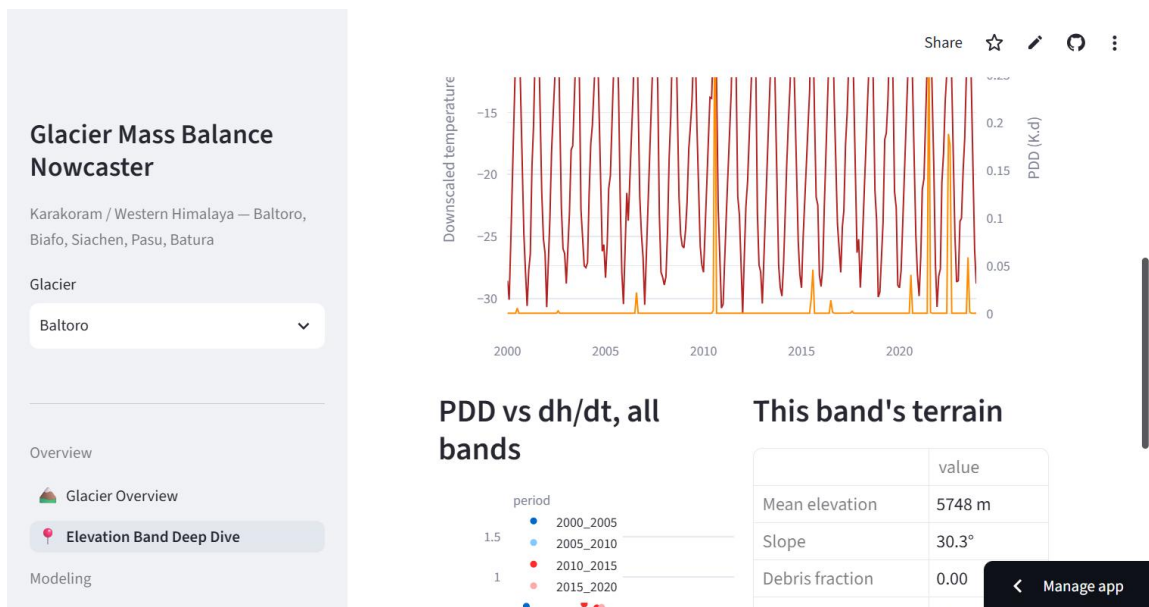


Figure 12: Elevation Band Deep Dive page, showing detail for Baltoro’s 5,700 m band.

7.3 Predicted Mass Balance

The final page runs the trained PINN’s nowcast for the selected glacier, reporting the nowcast value, the 2000–2020 observed average, the change between them, and the model’s historical holdout RMSE, alongside an elevation profile comparing observed history against the nowcast directly.

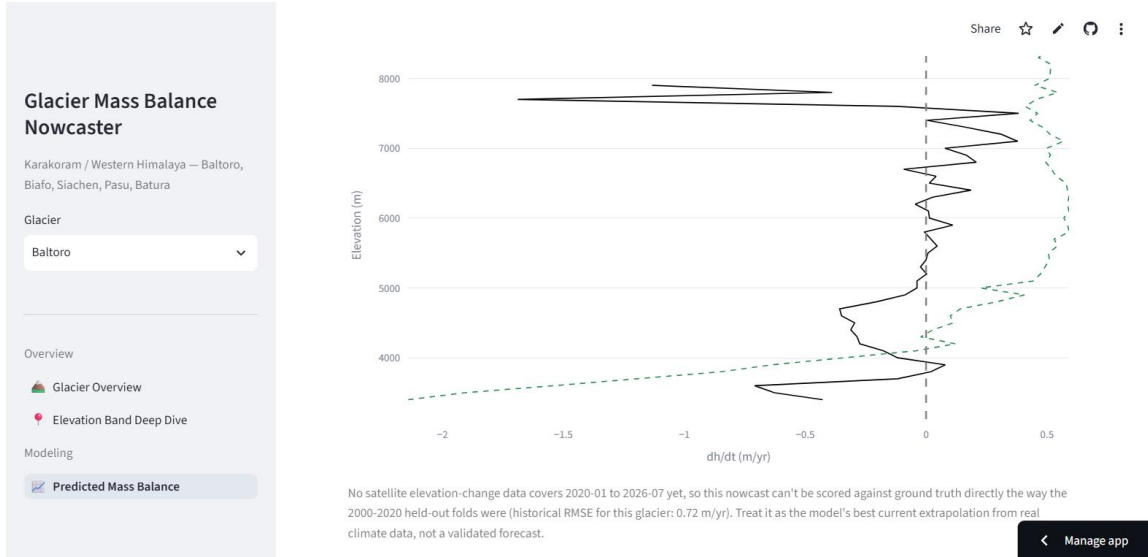


Figure 13: Predicted Mass Balance page, showing Baltoro’s PINN nowcast against its 2000–2020 observed profile.

7.4 Accompanying Article

Alongside this technical report, the project’s findings are presented in a long-form article for a general audience, published on Medium as *“Nowcasting Karakoram Glacier Mass Balance With a Physics-Informed Neural Network”*. The article walks through the project in the order the findings actually arrived, including the causal result that did not survive a more rigorous test and the ablation result showing the physics-loss term does not clearly improve accuracy, explained in accessible language for a general audience, with a direct link to the live dashboard.

8 Ethics and Limitations

This model has not been operationally validated; its nowcast should not be used to inform water-resource or policy decisions without independent validation. Several specific limitations warrant discussion.

First, the nowcast has not been checked against satellite ground truth, since none exists yet for the post-2020 window; it should be read as the trained model’s best extrapolation from observed climate data, not as a validated forecast. A basin-scale plausibility check against GRACE’s own trend over the same window is a natural next step that has not yet been carried out.

Second, the ablation study’s central finding, that the physics-melt term does not clearly improve held-out accuracy, is specific to this data and this particular physical baseline. This does not imply that physics-informed losses are ineffective for this problem in general; it indicates that anchoring to a baseline already shown to be biased at low elevation is net detrimental more often than beneficial, and a smoothly saturating melt formulation is a concrete, untested remedy.

Third, the debris-cover dataset originates from a published source, but it is built on an older glacier-outline standard (RGI 6.0) than the one used elsewhere in this project (RGI 7.0), reconciled by matching on glacier name and geometry instead of a shared identifier;

as noted in Section 5.2, the partial-correlation results have not been recomputed with this dataset. Precipitation is not elevation-corrected the way temperature is, and ERA5-Land is known to underestimate high-altitude snowfall in this specific range. Finally, with roughly 700 training rows per holdout fold, run-to-run variance from random initialization alone is on the order of a third of the typical mass-balance signal, so single-run RMSE comparisons in this report should not be read as precise to the third decimal.

9 Conclusion

This project develops a physics-informed neural network for Himalayan glacier mass balance whose physics-loss weights are mechanically derived from a preceding causal analysis. The physics-melt term does not clearly earn its place on held-out accuracy, and the PINN does not surpass XGBoost on raw RMSE; its concrete advantage is that its predicted elevation profiles are measurably smoother and more physically coherent than either baseline. Extending the validated model past the satellite record produces an operational nowcast, exposed through a live dashboard alongside the rest of the pipeline. Future work includes a smoothly saturating melt formulation to test whether the physics term can be improved, a basin-scale plausibility check against GRACE over the nowcast window, and recomputing the causal analysis with the Scherler debris-cover dataset.

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